

Using Sets and Venn Diagrams to Organize Real-World Data

Solutions

Math 140 – Math for Elementary Teachers

Part 1: Start by Hand

For the sample of 10 observations,

$$A = \{\text{observations from burned sites}\}, \quad B = \{\text{observations from grassland sites}\}.$$

1. Observations that belong to A :

$$A = \{1, 3, 5, 7, 8, 10\}$$

2. Observations that belong to B :

$$B = \{1, 2, 5, 6, 8\}$$

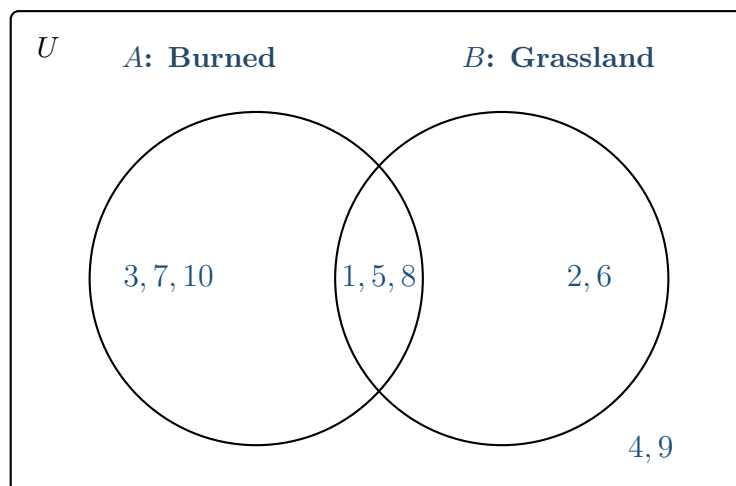
3. Observations that belong to $A \cap B$:

$$A \cap B = \{1, 5, 8\}$$

4. Observations that belong to neither A nor B :

$$\{4, 9\}$$

Completed Venn Diagram



Part 3: Two Sets Using the Full Dataset

For the complete dataset,

$$A = \text{Burned}, \quad B = \text{Grassland}, \quad n(U) = 123.$$

Set	Count
Total observations, $n(U)$	123
Burned observations, $n(A)$	72
Grassland observations, $n(B)$	37
Burned and Grassland, $n(A \cap B)$	23

Finding the regions

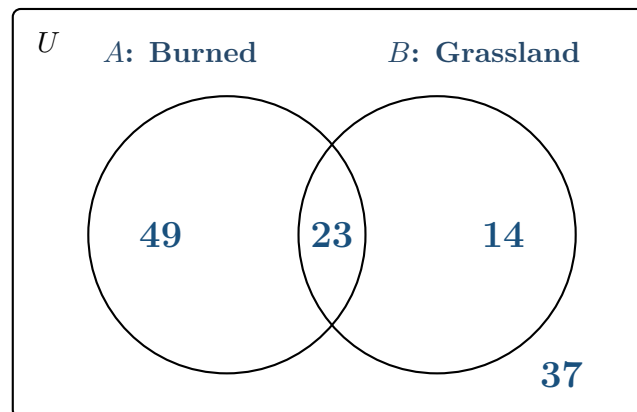
$$n(A \cap B) = \boxed{23}$$

$$A \text{ only : } 72 - 23 = \boxed{49}$$

$$B \text{ only : } 37 - 23 = \boxed{14}$$

$$\text{Neither : } 123 - (49 + 23 + 14) = \boxed{37}$$

Completed Venn Diagram



Step 6 Answers

1.

$$n(A \cap B) = 23$$

2.

$$n(A \cup B) = 49 + 23 + 14 = 86$$

3.

$$n(A') = 123 - 72 = 51$$

4.

$$n(B') = 123 - 37 = 86$$

5.

$$n(A \cup B') = 123 - 14 = 109$$

6.

$A \cap B$: observations that are both Burned and Grassland.

7.

$A \cup B'$: observations that are Burned or not Grassland, or both.

Part 4: Adding a Third Set

$A = \text{Burned}, \quad B = \text{Grassland}, \quad C = 2021.$

Set or Intersection	Count
$n(A)$: Burned	72
$n(B)$: Grassland	37
$n(C)$: 2021	68
$n(A \cap B)$: Burned and Grassland	23
$n(A \cap C)$: Burned and 2021	41
$n(B \cap C)$: Grassland and 2021	20
$n(A \cap B \cap C)$: Burned, Grassland, and 2021	12

Work from the center outward.

$$A \cap B \cap C = \boxed{12}$$

$$A \cap B \text{ only} = 23 - 12 = \boxed{11}$$

$$A \cap C \text{ only} = 41 - 12 = \boxed{29}$$

$$B \cap C \text{ only} = 20 - 12 = \boxed{8}$$

$$A \text{ only} = 72 - (11 + 29 + 12) = \boxed{20}$$

$$B \text{ only} = 37 - (11 + 8 + 12) = \boxed{6}$$

$$C \text{ only} = 68 - (29 + 8 + 12) = \boxed{19}$$

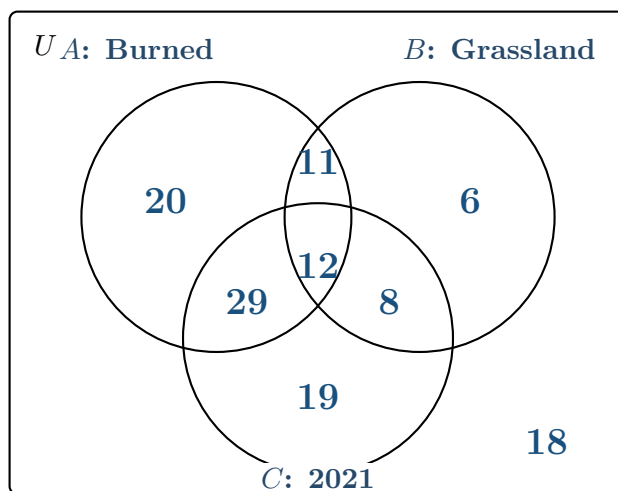
The total inside the three circles is

$$20 + 6 + 19 + 11 + 29 + 8 + 12 = 105.$$

Therefore,

$$\text{None} = 123 - 105 = \boxed{18}.$$

Completed Three-Set Venn Diagram



Part 4: Interpret Your Diagram

1. How many observations belong to all three sets?

12

2. How many observations are Burned and Grassland but not from 2021?

11

3. How many observations are Burned and from 2021 but not Grassland?

29

4. How many observations belong to none of the three sets?

18

5. $A \cap B \cap C$ represents

observations that are Burned, Grassland, and from 2021.

Part 5: Independent Application

There are 40 future elementary teachers.

$$n(M) = 24, \quad n(S) = 18, \quad n(M \cap S) = 10.$$

Start with the intersection:

$$M \cap S = 10$$

Mathematics only:

$$24 - 10 = 14$$

Science only:

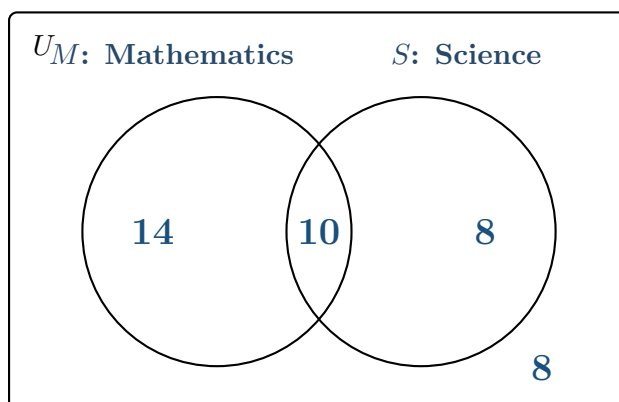
$$18 - 10 = 8$$

Inside the two circles:

$$14 + 10 + 8 = 32$$

Neither:

$$40 - 32 = 8$$



Mathematics only = 14,
 Science only = 8,
 Neither = 8,
 $n(M \cup S) = 32$.

Final Reflection – Sample Responses

1. How did a Venn diagram help organize the information?

The Venn diagram separated the observations into groups and showed which observations shared more than one characteristic.

2. What did Excel make easier?

Excel made it faster to sort and count the 123 observations without checking every row by hand.

3. Why was it still important to construct the diagrams by hand?

Constructing the diagrams by hand helped show how the counts fit into the different regions and reinforced the mathematics behind the sets.

4. Give another real-world use for sets or Venn diagrams.

Answers will vary. Examples include organizing student survey responses, comparing school activities, classifying animals, or comparing characteristics of numbers or shapes.

Confidence question:

Answers will vary.

One thing I understand better now is:

Answers will vary.